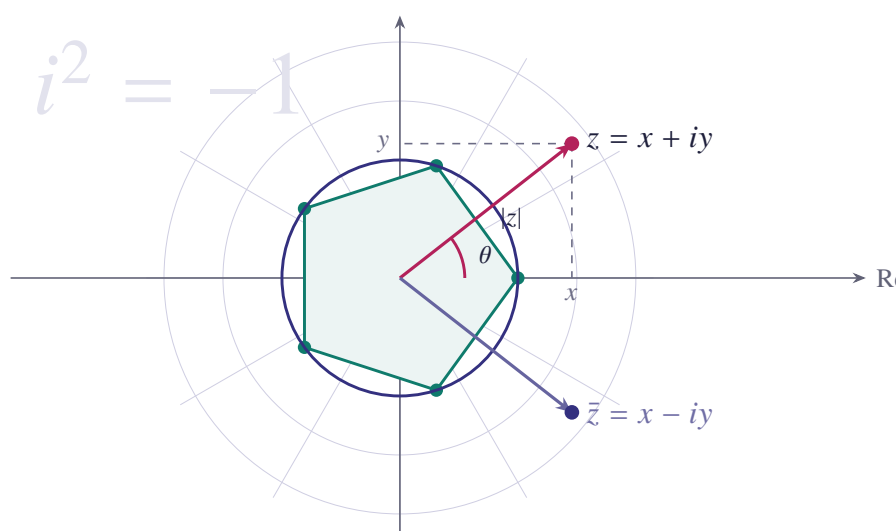


Complex Numbers

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8.1 The Set of Complex Numbers

Definition 8.1 | Complex numbers

The set of complex numbers generalizes the real numbers on the real line to a plane of numbers having two components, a real one and an imaginary one.

$$\mathbb{R} \rightarrow \mathbb{C}$$

$$\mathbb{C} = \{(x, y) \mid x, y \in \mathbb{R}\}, \quad z = (x, y), \quad x = \operatorname{Re}\{z\}, \quad y = \operatorname{Im}\{z\}$$

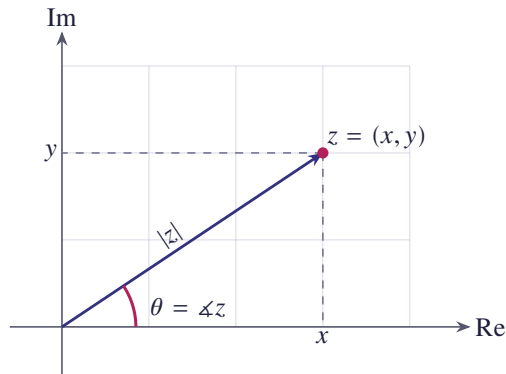


Figure 8.1 The complex number $z = (x, y)$ in the complex plane; its modulus $|z|$ and argument θ

Magnitude or norm of a complex number:

$$|z| = \sqrt{x^2 + y^2}$$

Phase or argument:

$$\theta = \angle z = \tan^{-1}\left(\frac{y}{x}\right), \quad z = |z| \angle \theta$$

$$x = |z| \cos \theta, \quad y = |z| \sin \theta$$

$$z = x \times (1, 0) + y \times (0, 1) = x + iy$$

★ Key result 8.2

$$i^2 = -1$$

8.2 The Four Basic Operations

Let $z_1 = (a_1, b_1) = a_1 + ib_1$ and $z_2 = (a_2, b_2) = a_2 + ib_2$.

(I) Addition:

$$z_1 + z_2 = (a_1 + a_2, b_1 + b_2) = a_1 + a_2 + i(b_1 + b_2)$$

(II) Subtraction:

$$z_1 - z_2 = (a_1 - a_2, b_1 - b_2) = a_1 - a_2 + i(b_1 - b_2)$$

(III) Multiplication:

$$\begin{aligned} z_1 \times z_2 &= (a_1 + ib_1) \times (a_2 + ib_2) \\ &= a_1 \times a_2 + a_1 \times ib_2 + ib_1 \times a_2 + ib_1 \times ib_2 \\ &= (a_1 a_2 - b_1 b_2) + i(a_1 b_2 + b_1 a_2) \end{aligned}$$

Multiplication by a real number:

$$\forall \alpha \in \mathbb{R} : \quad \alpha z_1 = \alpha a_1 + i \alpha b_1$$

(IV) Division:

$$\begin{aligned} z_1 \div z_2 &= (a_1 + ib_1) \div (a_2 + ib_2) = \frac{a_1 + ib_1}{a_2 + ib_2} = \frac{(a_1 + ib_1)(a_2 - ib_2)}{(a_2 + ib_2)(a_2 - ib_2)} \\ &= \frac{(a_1 a_2 + b_1 b_2) + i(b_1 a_2 - a_1 b_2)}{a_2^2 + b_2^2} = \frac{a_1 a_2 + b_1 b_2}{a_2^2 + b_2^2} + i \frac{b_1 a_2 - a_1 b_2}{a_2^2 + b_2^2} \end{aligned}$$

8.3 Properties of Addition and Multiplication

Theorem 8.3 | The complex numbers form a field

The set of complex numbers, together with addition and multiplication, forms a field that cannot be ordered.

For $z_1, z_2, z_3 \in \mathbb{C}$:

$$z_1 + z_2 \in \mathbb{C}, \quad z_1 \times z_2 \in \mathbb{C}$$

$$z_1 + z_2 = z_2 + z_1, \quad z_1 \times z_2 = z_2 \times z_1$$

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3), \quad (z_1 z_2) z_3 = z_1 (z_2 z_3)$$

$$z_1 (z_2 + z_3) = z_1 z_2 + z_1 z_3$$

$$z_1 + (0, 0) = z_1 + 0 = z_1, \quad z_1 + (-z_1) = (0, 0) = 0$$

$$-z_1 = -(a_1, b_1) = (-a_1, -b_1) = -a_1 - ib_1$$

$$z_1 \times (1, 0) = z_1 \times 1 = z_1, \quad z_1 \times z_1^{-1} = (1, 0) = 1$$

$$z_1^{-1} = \left(\frac{a_1}{a_1^2 + b_1^2}, \frac{-b_1}{a_1^2 + b_1^2} \right) = \frac{a_1}{a_1^2 + b_1^2} - i \frac{b_1}{a_1^2 + b_1^2}$$

Remark | No ordering

An order relation $z_1 < z_2$ cannot be defined on the set of complex numbers.

8.4 The Complex Conjugate

Definition 8.4 | Complex conjugate of a number

$$z = x + iy \quad \Rightarrow \quad \bar{z} \triangleq x - iy$$

Example 8.5

Plot the conjugate of the complex number $z = 5 + 2i$ in the complex plane.

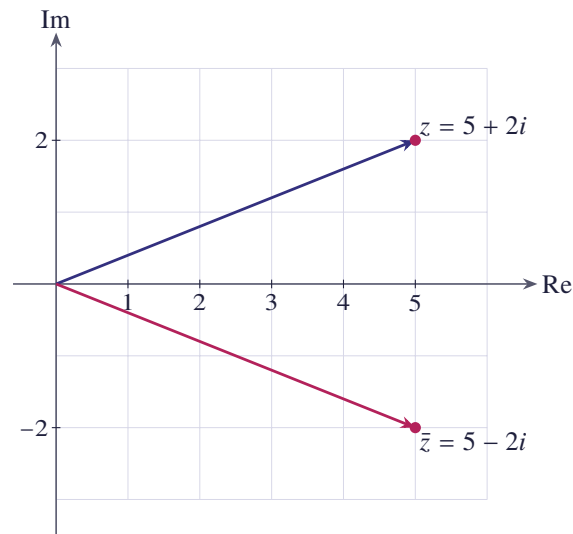


Figure 8.2 The complex number $z = 5 + 2i$ and its conjugate $\bar{z} = 5 - 2i$

Exercise 8.6

Prove the following statements.

$$\operatorname{Re}\{z\} = \frac{1}{2}(z + \bar{z}), \quad \operatorname{Im}\{z\} = \frac{1}{2i}(z - \bar{z}), \quad |z|^2 = z \cdot \bar{z}$$

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$$

$$|z| = |\bar{z}|, \quad |\operatorname{Re}\{z\}| \leq |z|, \quad |\operatorname{Im}\{z\}| \leq |z|$$

8.5 The Triangle Inequality

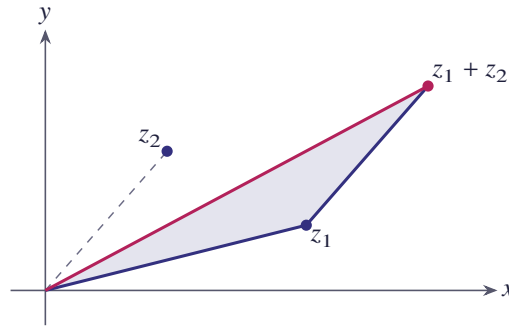


Figure 8.3 The triangle inequality: a triangle with vertices 0, z_1 and $z_1 + z_2$

Exercise 8.7

- Prove the triangle inequality for any two complex numbers: $|z_1 + z_2| \leq |z_1| + |z_2|$.
- Prove the general form of the triangle inequality for n arbitrary complex numbers:

$$|z_1 + z_2 + \cdots + z_n| \leq |z_1| + |z_2| + \cdots + |z_n|$$

- Prove that $|z_1 + z_2| \geq |z_1| - |z_2|$.

8.6 The Polar Form of Complex Numbers

$$z = x + iy, \quad \begin{cases} x = |z| \cos \theta \\ y = |z| \sin \theta \end{cases} \Rightarrow z = |z| (\cos \theta + i \sin \theta)$$

$$\theta = \text{Arg}(z) = \tan^{-1} \frac{y}{x}, \quad -\pi < \theta \leq \pi$$

$$\arg(z) = \text{Arg}(z) + 2k\pi, \quad k \in \mathbb{Z}$$

8.7 Multiplying Two Complex Numbers in Polar Form

$$z_1 = |z_1|(\cos \theta_1 + i \sin \theta_1), \quad z_2 = |z_2|(\cos \theta_2 + i \sin \theta_2)$$

$$\begin{aligned} z_1 \cdot z_2 &= |z_1||z_2|(\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) \\ &= |z_1||z_2|[(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) \\ &\quad + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)] \end{aligned}$$

★ Key result 8.8

$$z_1 \cdot z_2 = |z_1||z_2|[\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

 Remark | Consequence 1

When two complex numbers are multiplied, their magnitudes are multiplied and their phases are added.

 Remark | Consequence 2

If a complex number is raised to a natural power, its magnitude is raised to that power and its argument (phase) is multiplied by it (prove by induction). In other words:

$$z^k = |z|^k (\cos k\theta + i \sin k\theta)$$

$$z^k = [|z|(\cos \theta + i \sin \theta)]^k = |z|^k (\cos \theta + i \sin \theta)^k = |z|^k (\cos k\theta + i \sin k\theta)$$

★ **Key result 8.9** | De Moivre's formula

$$(\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$$

 Exercise 8.10

Using De Moivre's formula, express $\cos 3\theta$ and $\cos 4\theta$ in terms of $\cos \theta$ and $\sin \theta$.

8.8 The m -th Root of a Complex Number

$$w = \sqrt[m]{z} = z^{1/m}, \quad z = |z|(\cos \theta + i \sin \theta)$$

$$z = w^m, \quad w = |w|(\cos \varphi + i \sin \varphi)$$

$$z = |z|(\cos \theta + i \sin \theta) = |w|^m (\cos m\varphi + i \sin m\varphi)$$

$$|w|^m = |z| \Rightarrow |w| = \sqrt[m]{|z|}$$

$$m\varphi = \theta + 2k\pi \Rightarrow \varphi = \frac{\theta + 2k\pi}{m}, \quad k = 0, 1, 2, \dots, m-1$$

★ **Key result 8.11**

$$w = \sqrt[m]{z} = \sqrt[m]{|z|} \left[\cos \left(\frac{\theta + 2k\pi}{m} \right) + i \sin \left(\frac{\theta + 2k\pi}{m} \right) \right], \quad k = 0, 1, 2, \dots, m-1$$

Example 8.12

What are the roots of the equation $z^n = 1$?

$$z = \sqrt[n]{1}, \quad 1 = 1 \angle 0 \Rightarrow z = \cos\left(\frac{2k\pi}{n}\right) + i \sin\left(\frac{2k\pi}{n}\right), \quad k = 0, 1, \dots, n-1$$

Defining $\omega = 1 \angle \frac{2\pi}{n}$, the roots are

$$z = 1, \omega, \omega^2, \dots, \omega^{n-1}$$

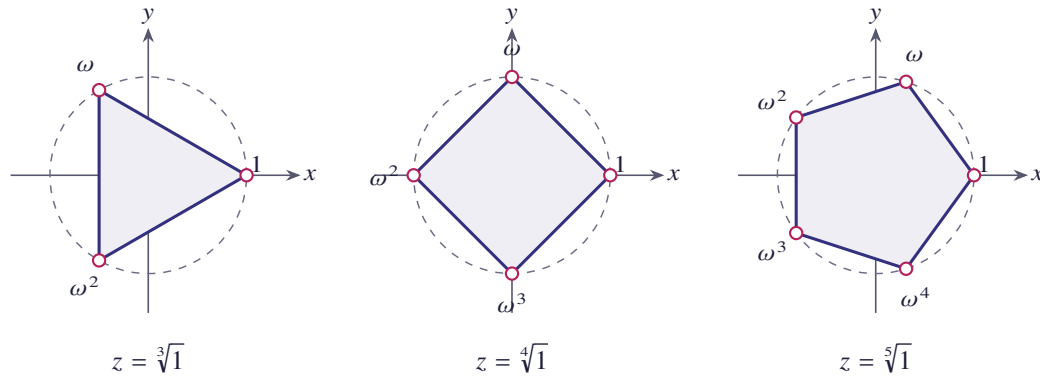


Figure 8.4 The roots $z = \sqrt[3]{1}$, $z = \sqrt[4]{1}$ and $z = \sqrt[5]{1}$ on the unit circle

Exercise 8.13

- Find the roots of the equation $z^6 + 6 + 8i = 0$ and plot them in the complex plane.
- Verify the identity $1 + z + z^2 + \dots + z^n = \frac{1 - z^{n+1}}{1 - z}$ and then derive the following identity (Lagrange's identity).

$$1 + \cos \theta + \cos 2\theta + \dots + \cos n\theta = \frac{1}{2} + \frac{\sin\left[\left(n + \frac{1}{2}\right)\theta\right]}{2 \sin \frac{\theta}{2}}$$